

Four problems — answers

2 — The same mean, made in bursts

$$\text{Fano} \approx 1 + b = 5, \quad \eta^2 = \frac{5}{25} = 0.20, \quad \eta = 0.447.$$

More than twice item 1's 0.20, at the same mean. **What changed that the mean cannot see** is the size of each arrival. Item 1's proteins arrive one at a time; these arrive four at a time, from a quarter as many transcripts. The mean is the product of how often and how many, so it cannot tell the two apart. The spread can.

The design rule this gives you: at a fixed mean, a strong promoter with a weak RBS is quieter than a weak promoter with a strong RBS. A promoter library and an RBS library that reach the same expression level are not interchangeable.

Which to build for 25 per cell with the least variation: raise the promoter and weaken the RBS until b is near 1. At $b = 1$ the Fano factor is about 2 and $\eta = \sqrt{2/25} = 0.28$; the floor, as b falls toward zero, is the Poisson's 0.20. Going lower than that takes feedback, which is item 3.

The exact two-stage result is $\text{Fano} = 1 + k_p/(\gamma_m + \gamma_p)$, which is $1 + b$ only when the mRNA lifetime is much shorter than the protein's. In session 12's simulation, $b = 10$ with mRNA ten times shorter-lived gave an exact Fano factor of 10.09, against $1 + b = 11$.

3 — The same gene, repressing itself

$$g = \frac{3 \times 1}{1 + 1} = 1.5, \quad \text{Fano} = \frac{1}{1 + 1.5} = 0.4$$

$$\eta_{\text{constitutive}} = \frac{1}{\sqrt{100}} = 0.100, \quad \eta_{\text{self-repressed}} = \sqrt{\frac{0.4}{100}} = 0.063$$

Cutting the Fano factor to 0.4 cuts η by $\sqrt{2.5} = 1.58$. The slide showed $n = 1, 2, 4$ and 8; $n = 3$ falls between them, as it should.

What the feedback changed. Not the event rate: at the same mean, births and deaths each happen at γx per unit time with or without feedback, so the noise going *in* is identical. What changed is how fast a fluctuation is pulled back. A cell that drifts high makes less, so it returns at rate $\gamma(1 + g)$ instead of γ . Each fluctuation lasts a shorter time, and the variance is the noise going in divided by twice that return rate. It is the same mechanism that made the self-repressed gene respond faster in session 7.

Two limits, both worth saying. The result is for proteins made one at a time; with bursts the benefit is smaller. And the result is about intrinsic noise only. Slow extrinsic changes in the gene's own rates are

buffered too, and more strongly: at steady state $\gamma x = f(x)$, so $d \ln x^* / d \ln \beta = 1 / (1 + g)$, and a slow wobble in the promoter strength reaches x divided by $1 + g$ in CV, not by $\sqrt{1 + g}$.

4 — Reading Table 1

There is more than one good answer. Three comparisons that each tell you something:

- **RP22 with and without IPTG.** Same strain, inducer only. Intrinsic noise falls from 25 to 6.3 as expression rises about thirtyfold (intensity $0.030 \rightarrow 1.00$). That is the $1/\langle n \rangle$ law: more transcripts, smaller relative spread. Extrinsic noise falls too, from 33 to 9.8, because saturating IPTG makes the cell-to-cell variation in LacI irrelevant (the paper's own explanation, note 13, p. 1186).
- **M22 against M22 + repressilator.** Extrinsic noise jumps from 5.4 to 42. The clock moves LacI up and down in every cell, out of phase between cells, so both promoters swing together — exactly what extrinsic means. Intrinsic noise is 12, but the mean is lower too (intensity 0.18), so compare it with the paper's own fit $\eta_{\text{int}}^2 \approx c_1/m + c_2$, which predicts about 8 at that mean. About 1.5 times higher: noise is larger while a gene is being switched than at a steady state (p. 1186).
- **M22 against RP22.** The constitutive strain is the quiet baseline the paper starts from: about 5% each way.

The check. RP22: $25^2 + 33^2 = 1714$, and $\sqrt{1714} = 41.4$, against 41 in the table. RP22 + IPTG: $6.3^2 + 9.8^2 = 135.7$, $\sqrt{135.7} = 11.65$, against 11.7. The table rounds; the identity holds.